0393

National University of Computer and Emerging Sciences

FAST School of Computing — Fall 2022 — Islamabad Campus

Serial No:

MT-1004 Linear Algebra

BS(CS)

Wednesday, 9th November, 2022

Course Instructors

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Sessional Exam-II

Total Time: 1 Hour

Total Marks: 80

Signature of Invigilator

Student Name | Roll No | Section | Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work. Use the back of the last page for rough work.
3. If you need more space write on the back side of the paper and clearly mark question and part number etc.
4. After asked to commence the exam, please verify that you have 10 pages different printed pages including this title page. There are a total of 3 questions.
5. Calculator sharing is strictly prohibited.
6. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.
7. Fit in all your answers in the provided space. You may use extra space on the last page if required. If you do so, clearly mark question/part number on that page to avoid confusion.

	Q-1	Q-2	Q-3	Total
Marks Obtained				
Total Marks	35	20	25	80

Solution

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Question # 1 [35 marks]

(a) In $\mathbb{P}_2$, find the change-of-coordinates matrix $P_{\mathbb{C}\leftarrow\mathbb{B}}$ from the basis $\mathbb{B} = \{\underbrace{1-2t+t^2}_{b_1}, \underbrace{3-5t+4t^2}_{b_2}, \underbrace{2t+3t^2}_{b_3}\}$ to the standard basis $\mathbb{C} = \{\underbrace{1}_{c_1}, \underbrace{t}_{c_2}, \underbrace{t^2}_{c_3}\}$. Moreover find the coordinate vector for $-1+2t$ w.r.t $\mathbb{B}$ basis. [10]

$$b_1 = x_1 c_1 + y_1 c_2 + z_1 c_3 \Rightarrow 1-2t+t^2 = 1c_1 - 2c_2 + 1c_3$$

$$b_2 = x_2 c_1 + y_2 c_2 + z_2 c_3 \Rightarrow 3-5t+4t^2 = 3c_1 - 5c_2 + 4c_3$$

$$b_3 = x_3 c_1 + y_3 c_2 + z_3 c_3 \Rightarrow 2t+3t^2 = 0c_1 + 2c_2 + 3c_3$$

$$P_{C\leftarrow B} = \left[[b_1]_C \quad [b_2]_C \quad [b_3]_C \right]$$

$$= \begin{bmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$$

$$= (P_{C\leftarrow B})^{-1} = \begin{bmatrix} -23 & -9 & 6 \\ 8 & 3 & -2 \\ -3 & -1 & 1 \end{bmatrix}$$

$$x = -1+2t$$

$$[x]_B = P_{B\leftarrow C} [x]_C$$

$$= \begin{bmatrix} -23 & -9 & 6 \\ 8 & 3 & -2 \\ -3 & -1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 23-18 \\ -8+6 \\ 3-2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

(b) Find the basis for following subspaces [5+5]

(i) span$\{1-2x, x-x^2, 1-x^2, 1-2x+x^2\}$ in $\mathbb{P}_2(\mathbb{R})$

$$\simeq \left\{ \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right\}$$

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ -2 & 1 & 0 & -2 \\ 0 & -1 & -1 & 1 \end{pmatrix} \underset{\sim}{R_2+2R_1} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & -1 & -1 & 1 \end{pmatrix}$$

$$\underset{\sim}{R_3+R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$ basis are $\{1-2x, x-x^2, 1-x^2\}$

(ii) span $\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \right)$

$$\simeq \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 1 & 0 & -1 & 1 \end{pmatrix} \underset{\sim}{R_4-R_1} \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\underset{\sim}{R_3-R_2} \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Basis are $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$

(c) Let $T : \mathbb{R}^3 \to \mathbb{P}_3(\mathbb{R})$ be a linear transformation such that [5+5+5]

$$T\begin{bmatrix} a \\ b \\ c \end{bmatrix} = (a+b)x^3 + bx^2 + cx$$

Find

(i) Matrix of transformation

$$T\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 1x^3 + 0x^2 + 0x \simeq \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$T\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 1x^3 + 1x^2 + 0x \simeq \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$T\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0x^3 + 0x^2 + 1x \simeq \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

Matrix of transformation $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

(ii) $ker(T)$ or $Nullspace(A)$

Since all columns are pivot

$$\Rightarrow ker(T) = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

(iii) $Range(T)$ or $Columnspace\ of\ matrix\ of\ T$

For range of T $\left(\begin{array}{ccc|c} 1 & 1 & 0 & x \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & w \end{array}\right)$

is consistent if $w = 0$ i.e

$$Range(T) = \{ xt^3 + yt^2 + zt + w \mid w = 0 \}$$

Question # 2[20 marks]

(a) Give short answer but it MUST contain proper reasoning [4+3]

(i) If v is an eigenvector of A with corresponding eigenvalue λ and c is a scalar. Find an eigenvector and an eigenvalue of $A - cI$ and A^{-1}.

$Av = \lambda v$

$(A-cI)v = (\lambda - c)v$ $\Rightarrow$ eigenvalues of $A-\lambda I$ is $\lambda - c$, & eigenvector is v

$A^{-1}v = \frac{1}{\lambda} v$ $\Rightarrow$ eigenvalue of A^{-1} is $\frac{1}{\lambda}$ eigenvector is v.

(ii) Construct a 3×3 matrix that has eigenvalues 0, -1 and 5. Comment about the invertibility of constructed matrix.

Since 0 is eigenvalue of given matrix

$\Rightarrow$ it is non invertible.

(b) If a 2×2 matrix A has eigenvalues 2 and 3, then [2+2]

(i) Write the characteristic polynomial of A.

$\lambda^2 - tr(A)\lambda + det(A) = 0$

where $tr(A) = \lambda_1 + \lambda_2 = 5$

$det(A) = \lambda_1 \lambda_2 = 6$

Characteristic polynomial $\lambda^2 - 5\lambda + 6 = 0$

(ii) If $A \sim B$, i.e, $A = CBC^{-1}$ for some invertible matrix C. Determine the $tr(B)$ and $det(B)$.

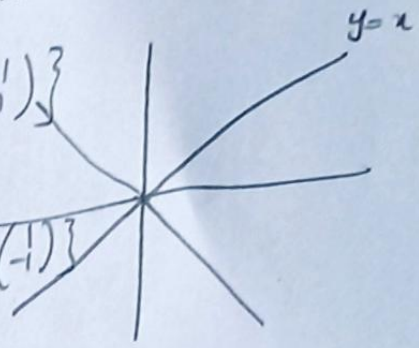
Since $A \sim B$

$\Rightarrow tr(A) = tr(B) = 5$

$det(A) = det(B) = 6$.

(c) Find the eignvalues and corresponding eigenspaces (by Inspection, Without Finding Matrix of Transformation) of the following transformation in $\mathbb{R}^2$. [3+3+3]

(i) Reflection about the line $y = x$.



1) all vector on $y = x$ i.e $\text{span}\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\}$ with eigenvalue 1.

2) all vectors on $y = -x$ i.e $\text{span}\left\{\begin{pmatrix}1\\-1\end{pmatrix}\right\}$ with eigenvalue -1.

(ii) Projection on the x-axis .

(i) all vectors on x-axis i.e $\text{span}\left\{\begin{pmatrix}1\\0\end{pmatrix}\right\}$ with eigenvalue 1

(ii) all vector on y-axis i.e $\text{span}\left\{\begin{pmatrix}0\\1\end{pmatrix}\right\}$ with eigenvalue 0.

(iii) Contraction with factor $k(0 \leq k < 1)$.

all vectors of $\mathbb{R}^2$ are eigenvectors with eigenvalue k.

Question #3[25 marks]

[6+4+2+4]

(a) For the given matrix

$$A = \begin{bmatrix} -4 & 2 \\ 3 & 1 \end{bmatrix}, \quad \text{and vector} \quad x_0 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

Find

(i) The eigenvalues and corresponding eigenspaces

$$\lambda^2 + 3\lambda - 10 = 0$$

$$\lambda^2 + 5\lambda - 2\lambda - 10 = 0$$

$$\lambda = -5 \quad \lambda = 2$$

-5 eigenspace $\rightarrow$ span $\left\{ \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\}$

$$[A+5I \mid 0] \sim \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 6 & 0 \end{array}\right]$$

2 eigen s/pace $\rightarrow$ span $\left\{ \begin{pmatrix} 1/3 \\ 1 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right\}$

$$[A-2I \mid 0] \sim \left[\begin{array}{cc|c} -6 & 2 & 0 \\ 3 & -1 & 0 \end{array}\right]$$

(ii) $A^{10}x_0$.

$$A^{10}x_0 = c_1 \lambda_1^{10} v_1 + c_2 \lambda_2^{10} v_2$$

$$= (-5)^{10} \begin{pmatrix} -2 \\ 1 \end{pmatrix} + 2^{10} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} -2 & 1 & -1 \\ 1 & 3 & 4 \end{array}\right)$$

$$R_2 + 2R_1 \left(\begin{array}{cc|c} 1 & 3 & 4 \\ 0 & 7 & 7 \end{array}\right)$$

$c_2 = 1, \; c_1 = 1$

(iii) $A^k x_0$. What happens as k becomes large (i.e.. $k \to \infty$)? Explain.

$$A^k x_0 = (-5)^k \begin{pmatrix} -2 \\ 1 \end{pmatrix} + 2^k \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

as $k \to \infty$ $\quad A^k x_0 \to \infty$

$\Rightarrow$ origin is repeller.

(iv) The solution of the difference equation $x_{k+1} = Ax_k$.

The solution of $x_{k+1} = Ax_k$

$k=0$ $\quad x_1 = Ax_0$

$k=1$ $\quad x_2 = A^2 x_0$

$\vdots$

$k=n$ $\quad x_n = A^n x_0$

$= c_1 \lambda_1^n v_1 + c_2 \lambda_2^n v_2$

$= (-5)^n \begin{pmatrix} -2 \\ 1 \end{pmatrix} + (2)^n \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

(b) Find the solution of the following differential system (by using the notion of eigenvalues and eigenvectors) [6]

$$\frac{dx_1}{dt} = -4x_1 + 2x_2, \quad \frac{dx_2}{dt} = 3x_1 + x_2,$$

subject to initial condition $x_1(0) = -1$ and $x_2(0) = 4$. Also find the steady state solution of the system.

Notice: Coefficient matrix is same as given in part (a), so avoid unnecessary repetition of computations.

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} -4 & 2 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} -5 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 1 & 3 \end{pmatrix}^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = PDP^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Let $P^{-1}X = Y$

$\Rightarrow Y' = DY$

$$\begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} -5 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

$\frac{du}{dt} = -5u \Rightarrow u = c_1 e^{-5t}$ $\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = c_1 v_1 e^{-5t} + c_2 v_2 e^{2t}$

$\frac{dv}{dt} = 2v$ $\quad v = c_2 e^{2t}$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = c \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-5t} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{2t}$$

(c) Solve the following recurrence relation [3]

$$x_n = 4x_{n-1} - 3x_{n-2}$$

subject to the initial conditions $x_0 = 0,\ x_1 = 1$.

$\lambda^2 - 4\lambda + 3 = 0$

$\lambda^2 - 3\lambda - \lambda + 3 \ 0$

$\lambda = 3,\ \lambda = 1$

$x_n = C_1 \lambda_1^n + C_2 \lambda_2^n$

$= C_1 3^n + C_2 (1)^n$

$n = 0$

$x_0 = C_1 + C_2 \Rightarrow 0 = C_1 + C_2$

$n = 1$

$x_1 = 3C_1 + C_2 \Rightarrow 1 = 3C_1 + C_2$.

$C_1 = 1/2$

$C_2 = -1/2$

$\Rightarrow x_n = \frac{1}{2} 3^n - \frac{1}{2}$